

CODE SPORT



In this month's column, we continue our discussion of natural language processing, focusing on the topic of abbreviation expansion, which is particularly important for processing medical texts.

Over the past few months, we have been discussing natural language processing (NLP) algorithms. In this month's column, we continue our discussion on NLP by focusing on abbreviation expansion, specifically in medical texts processing. Many texts contain abbreviations which are a short representation of a long phrase. Typically, abbreviations are formed by using the first letter of the multiple words that constitute the expanded phrase. For instance, IBM stands for International Business Machines or IPL for Indian Premier League.

A single abbreviation can potentially match multiple expanded phrases. For instance, consider the abbreviation 'WWF'. It can expand to 'World Wildlife Fund' or 'World Wrestling Federation'. Given a piece of text which contains abbreviations, for each abbreviation, the NLP system needs to find the expansion that is the best matched candidate for that abbreviation. In that sense, abbreviation expansion can be considered similar to the problem of the word sense disambiguation problem in NLP.

In word sense disambiguation (WSD), given a word which can have multiple different meanings or senses, and the text which contains that word, the task is to identify the particular sense conveyed by that word in that specific context in the document. For example, consider the word 'bank', which can mean either a financial institution or the side of a river. Depending on the context of usage in the text, in many cases, it is possible for the NLP system to figure out what is the correct meaning or sense associated with the word in the document. Consider the two sentences: 'I deposited the cheque in the bank' and 'The water level near the left bank was quite shallow'. It is easy to figure out the correct sense for the word 'bank' in these two sentences. However, there can be situations in which it is

difficult to disambiguate between the different senses of a given word in a text.

Similar to WSD, abbreviation sense disambiguation needs to be performed to identify the best matching expansion for an abbreviation, based on the context. Just as in WSD, domain knowledge or knowledge about the category to which the document belongs, can help to disambiguate across different possible expansions. Consider the abbreviation 'CBI'. In a news article regarding crimes, the likely expansion would be Central Bureau of Investigation. On the other hand, if it appears in a report on the banking system in India, it is possible that CBI stands for Central Bank of India.

Abbreviation expansion in a text can be considered as a text normalisation process. It consists of three major steps. The first step is to identify all words which are abbreviations. The second step is to find potential expanded phrases that match each abbreviation. The third step is to process the candidate set, and identify the best matched expansion for each abbreviation.

Abbreviation identification is typically done in a text after identifying and removing all stop words. Typically, rule based pattern matching is performed to locate all the abbreviations in a given text. For instance, a simple rule would be to choose all words that are not found in a dictionary and which are formed by all capital letters. Potential candidate expansion phrases are usually found by analysing all the documents in the corpus, and identifying expanded phrases that could match that abbreviation. Typically, most documents contain the abbreviation co-located with the definition of the expanded phrase before the first use of the abbreviation.

While abbreviations are usually formed by using the first letter of each word in the expanded phrase,